$3R$

12 A

amplitude, $x_0 = \dfrac{2.00}{2} = 1.00 \text{ m}$

equilibrium position at $= 1.00$ m mark

at 0.75 m mark, $x = 0.75 - 1.00 = -0.25$ m
at 1.50 m mark, $x = 1.50 - 1.00 = 0.50$ m

If at $t = 0$, swing is at 0 m,
$x = -x_0 \cos \omega t$

$$x = -1.00 \cos\left(\frac{2\pi}{2.4} t\right)$$

At $x = -0.25$ m,

$$-0.25 = -1.00 \cos\left(\frac{2\pi}{2.4} t_1\right)$$

$t_1 = 0.5035$ s

At $x = 0.50$ m,

$$0.50 = -1.00\cos\left(\frac{2\pi}{2.4}t_2\right)$$

$$t_2 = 0.8000 \text{ s}$$

time taken, $t_2 - t_1 = 0.8000 - 0.5035 = 0.2965 = 0.30$ s

13 C